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Exploratory Data Analysis (EDA) Report

1. Introduction

Exploratory Data Analysis (EDA) is a crucial step in the data science pipeline used to understand the structure, patterns, and relationships within a dataset. The objective of this project is to analyze the given dataset, identify trends, detect anomalies, and extract meaningful insights using statistical techniques and visualizations.

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2. Dataset Description

The dataset used in this project contains multiple features (independent variables) and a target variable (dependent variable). It represents real-world data used to understand patterns and relationships between variables.

- **Dataset Name:** cleaned_data.csv
- **Number of Rows:** 463
- **Number of Columns:** 13
- **Target Variable:** price

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3. Data Understanding

3.1 Structure of Dataset

The dataset consists of both numerical and categorical features. Each row represents an observation, and each column represents a variable.

3.2 Data Types

- Numerical features: Continuous values such as age, salary, etc.
- Categorical features: Discrete values such as gender, category, etc.

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4. Data Cleaning

4.1 Missing Values

- Missing values were identified using `.isnull().sum()`.
- Rows with missing values were removed using `dropna()`.

4.2 Duplicate Values

- Duplicate rows were checked and removed if present.

4.3 Data Transformation

- Categorical variables were converted into numerical format using encoding techniques (e.g., one-hot encoding).

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5. Statistical Summary

Statistical measures such as mean, median, standard deviation, minimum, and maximum values were computed using `.describe()`.

Key Observations:

- The mean and median values indicate the central tendency of the data.
- Standard deviation shows the spread of the data.
- Some features show high variance, indicating possible outliers.

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6. Data Visualization

6.1 Histogram

Histograms were plotted to understand the distribution of numerical features.

- Most features follow a normal distribution.
- Some features are skewed.

6.2 Boxplot

Boxplots were used to detect outliers.

- Certain columns contain extreme values (outliers).
- Outliers may affect model performance.

6.3 Pairplot

Pairplots were used to visualize relationships between variables.

- Strong relationships were observed between some features.
- Some features show weak or no correlation.

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7. Correlation Analysis

A correlation matrix was plotted using a heatmap.

Key Findings:

- Strong positive correlation between certain features.
- Weak correlation between unrelated variables.
- Highly correlated features can influence the target variable significantly.

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8. Feature vs Target Analysis

Scatter plots were used to analyze the relationship between independent variables and the target variable.

Observations:

- Some features show a strong linear relationship with the target.
- Others show weak or no relationship.
- Important features influencing the target were identified.

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9. Key Insights

- The dataset contains [mention if clean or noisy].
- Some features have strong influence on the target variable.
- Outliers are present and may need further handling.
- Correlation analysis helped identify important features.
- Data visualization provided clear understanding of distributions and relationships.

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10. Conclusion

Exploratory Data Analysis helped in understanding the dataset effectively. Key patterns, relationships, and anomalies were identified. These insights are useful for building accurate machine learning models in the next stages.

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11. Future Work

- Perform feature selection based on correlation
- Handle outliers using advanced techniques
- Apply machine learning models for prediction
- Build dashboards for better visualization

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12. Tools & Technologies Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Google Colab

Final Outcome

This project successfully demonstrates the ability to analyze data, extract insights, and visualize patterns using EDA techniques.